

OKELLO ELLY

Software Developer | Systems Design & Architecture

Scalable Systems • Full-Stack Engineering • Backend Development • Cybersecurity • AI-Assisted Development

CONTACT INFORMATION

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PROFILE

Software Developer and Systems Architect with hands-on experience designing and building secure, scalable software solutions across digital learning, campus services, civic technology, analytics, and operational platforms. Skilled in full-stack development, backend engineering, REST API design, database architecture, mobile application development, networking, cybersecurity, and secure software development practices.

Experienced in applying software engineering principles including object-oriented programming, algorithm design, software testing, clean architecture, system design, and DevOps practices to develop reliable and maintainable applications. Strong foundation in backend systems, mobile development, computer networking, system configuration, and cybersecurity.

Founder and developer of technology-driven platforms including VumaGrid, SmartWaste, and Pioneer Africa, applying software architecture, data-driven engineering, and emerging technologies to solve real-world challenges.

AI-fluent developer experienced in AI-assisted development and workflow automation, leveraging tools such as Claude, Lovable, Base44, LangGraph, CrewAI, Zapier, and n8n to enhance software development, research, automation, and problem-solving.

TECHNICAL SKILLS

Programming Languages

Python, JavaScript, TypeScript, Java, C#, C++, C, SQL, HTML, Kotlin

Frontend Development

HTML, CSS, JavaScript, React, XML, Android UI Development

Backend Development

Python Flask, REST APIs, Backend Services, API Integration, JWT Authentication, RBAC

Mobile Development:

Android Studio, Kotlin, Java, XML layouts, mobile UI design, application architecture

Desktop & Visual Programming

C#, ASP.NET, Windows Forms

Databases

MySQL, SQLite, PL/SQL, Relational Database Design

Networking & Security

TCP/IP, VLANs, Routing, Switching, IP Addressing, Network Configuration, Linux Fundamentals, Cybersecurity Fundamentals, Secure System Design

Software Engineering

Object-Oriented Programming, Data Structures and Algorithms, Software Testing, Software Metrics, Quality Assurance, System Analysis, System Design, Software Architecture, DevOps Principles

Enterprise & Desktop Development:

C#, ASP.NET, Windows Forms, Console Applications

Tools

Git, GitHub, AI-Assisted Development Tools

PROFESSIONAL EXPERIENCE**Front-End AI Engineering Intern****FlyRank AI**

July 2026 – Present

- Develop frontend components and user interface improvements for AI-driven software products.
 - Collaborate with engineering teams on feature implementation, testing, documentation, and product iterations.
 - Apply modern frontend development practices to create maintainable and user-focused digital experiences.
 - Utilize AI-assisted development workflows to improve research, implementation efficiency, and technical problem solving.
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EDUCATION**Bachelor of Science in Computer Software Engineering****Zetech University**

January 2025 – November 2027

CORE TECHNICAL FOUNDATION :

- Software Development: Python, Java, JavaScript, TypeScript, C, C++, C#
- Object-Oriented Programming and Software Engineering Principles
- Full-Stack Web Development: HTML, CSS, JavaScript, React, REST APIs
- Backend Engineering: Flask, API Development, Authentication Systems, Database Integration
- Mobile Application Development: Kotlin, Java, Android Development, XML UI Design
- Visual Programming: C#, ASP.NET, Windows Forms, Desktop Applications
- Database Systems: MySQL, SQLite, PL/SQL, Database Design and Optimization
- Data Structures and Algorithm Design
- Computer Networking: TCP/IP, IP Addressing, VLANs, Routing, Switching, Network Configuration and Troubleshooting
- Cybersecurity and Secure Software Development
- Data Science and Analytics using Python
- System Analysis, System Design, and Software Architecture
- Software Testing, Software Metrics, Quality Assurance, and Engineering Practices
- System Configuration and Administration
- DevOps Concepts and Modern Software Delivery Practices

MAJOR ENGINEERING PROJECTS

VumaGrid — Founder & Systems Architect

AI-Powered Distributed Energy Management System (DERMS)

2026 – Present

- Founded and architecting an AI-powered Distributed Energy Management System (DERMS) designed to optimize energy usage in commercial buildings.
- Designing system architecture for monitoring, analytics, and intelligent decision-making across distributed energy resources.
- Applying AI concepts, data analytics, and software engineering principles to support energy forecasting, optimization, and operational efficiency.
- Developing the foundation for a scalable digital energy platform integrating renewable energy, storage systems, and building infrastructure.

SmartWaste — Founder & Software Developer

Smart Waste Management Platform

2026 – Present

- Founded and developed a digital waste management platform focused on improving waste collection visibility, tracking, and operational reporting.
- Designed backend systems and APIs supporting waste data collection, management, analytics, and reporting workflows.
- Developed database architectures supporting structured operational data and decision-making processes.

- Applied data-driven approaches to improve accountability, sustainability, and efficiency in waste management operations.
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BIT System — Backend Developer & System Architect

2024 – Present

Technologies: Python (Flask), MySQL, SQLite, JWT, REST APIs, HTML, CSS

- Designed and developed a campus digital platform supporting student services, event management, payments, analytics, and administrative workflows.
 - Implemented secure authentication using JWT and role-based access control (RBAC) for controlled system access.
 - Developed RESTful APIs enabling structured communication between system components.
 - Designed normalized relational database architectures improving scalability, maintainability, and data organization.
 - Applied software architecture principles to build modular and maintainable backend systems.
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Pioneer Africa — Founder & Technical Lead

Africa Rise Ecosystem

2026 – Present

- Founded a digital ecosystem focused on bridging gaps between learning, skills development, mentorship, and real-world opportunities for African students and young innovators.
 - Designed platform architecture supporting learning pathways, mentorship connections, innovation programs, and opportunity discovery.
 - Led product vision, technical direction, and ecosystem development strategy.
 - Developed technology-driven solutions aimed at improving access to skills growth and career opportunities.
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ADDITIONAL ENGINEERING PROJECTS

LearnLight — Software Developer

2026

- Contributed to the development of a digital learning platform supporting structured online education and accessible learning experiences.
- Developed frontend and backend components supporting content delivery and user interaction.

- Implemented modular architecture and API integrations to improve maintainability and scalability.
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Campus Pulse — Software Developer

2026

- Developed a university engagement platform connecting students with campus events, hackathons, sports tournaments, career fairs, and community activities.
 - Built features supporting event discovery, registration workflows, and student participation.
 - Designed systems improving communication between event organizers and students.
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Nairobi Hidden Gems — Software Developer

2026– Present

Technologies: HTML, CSS, JavaScript, Kotlin, XML, Android Development

- Developed a social discovery platform enabling users to explore aesthetic locations and experiences across Nairobi.
 - Expanded the platform into a mobile application using Kotlin and XML-based Android development.
 - Designed user-focused discovery features supporting location-based engagement and community exploration.
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My Sim League — Software Developer

2026 – Present

- Built a football management simulation platform providing a realistic managerial experience across Europe's top football leagues.
 - Designed gameplay systems, progression mechanics, and simulation logic.
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CERTIFICATIONS

Introduction to Modern AI — Cisco Networking Academy (2026)

- Artificial intelligence fundamentals
- Machine learning concepts
- AI applications

Junior Cybersecurity Analyst — Cisco Networking Academy

- Threat analysis
- Vulnerability assessment
- Security operations fundamentals

Huawei Computer Networks (CRA Training Program)

- Network architecture
- TCP/IP
- Routing and switching
- Network troubleshooting

Cisco Certified Cybersecurity Essentials

- Cyber risk management
- Secure access control
- Incident response

Data Science and Analytics — HP LIFE (2025)

- Data visualization
- Reporting and analytical decision-making

AI Fluency: Framework and Foundations — Anthropic (2025)

- Prompt engineering
- AI workflows
- AI-assisted software development

AI Fluency for Builders — Anthropic

Issued 2026

- AI delegation, AI-assisted development, responsible AI workflows, and developer productivity.

Teaching the AI Fluency Framework — Anthropic

Issued 2026

- AI fluency frameworks, AI literacy, and practical AI adoption.

Claude 101 — Anthropic

Issued 2026

- AI-assisted research, Claude workflows, and productivity applications.

AI Safari — Power Learn Project

Issued 2026

- Artificial intelligence applications, prompt engineering, AI agents, workflow automation, responsible AI, design thinking, ChatGPT, Claude, Gemini, Zapier, n8n, CrewAI, and LangGraph.

VOLUNTEER LEADERSHIP & PROFESSIONAL DEVELOPMENT

Aspire Leaders Program — Graduate, Cohort 1 (2026)

- Completed leadership development program focused on leadership, systems thinking, problem solving, and professional growth.

Aspire Leaders Program — Moderator, Cohort 2 and 3 (Volunteer) (2026)

- Facilitated participant discussions and supported community engagement.
- Assisted with program operations and peer learning activities.

Secretary — Innovation Club, Zetech University

- Supported innovation initiatives, technical activities, and student engagement programs.
- Coordinated communication, events, and project activities.

GMI Mentorship Program (2026)

- Participated in mentorship focused on leadership development and career growth.

Anticipatory Governance: Systems Thinking and Strategic Foresight (Parts 1 & 2) — 2026

- Systems thinking frameworks and strategic foresight implementation.

McKinsey Forward Program — 2026

- Professional development focused on problem solving, communication, leadership, and workplace effectiveness.

Cisco Networking Academy Learn-A-Thon Recognition — (2025& 2026)

- Recognized twice for participation and completion of Cisco learning activities supporting technology skills development